Dynamic Susceptibility continued

So, consider antiferromagnet.

① Choose operator $\hat{O} = \hat{S}^x_r \approx \sqrt{2S}\cdot\sqrt{\frac{2}{N}}\sum_k e^{-i\vec{k}\cdot\vec{r}}(a_k + a^+_{-k})$

Response along $\hat{x}$-axis too: $\hat{A} = \hat{S}^x_{r'}$

Then $\chi_{rr'}(\omega) = -i\int dt\, e^{i(\omega+i\delta)t}\langle[S^x_{r'}(t), S^x_r]\rangle$

$$= -i\int dt\, e^{i\omega t}(2S)\frac{2}{N}\sum_{kk'} e^{-i\vec{k}'\vec{r}'}e^{-i\vec{k}\vec{r}}\cdot$$

$$\cdot\langle[a_{k'}(t) + a^+_{-k'}(t), a_k + a^+_{-k}]\rangle$$

But eigenmodes are α_k and β_k:

$$a_k = ch\theta_k\,\alpha_k - sh\theta_k\,\beta^+_{-k}$$

$$b_{-k} = -sh\theta_k\,\alpha^+_k + ch\theta_k\,\beta_{-k}$$

$$a^+_{-k} = ch\theta_k\,\alpha^+_{-k} - sh\theta_k\,\beta_k \quad (\text{using } sh\theta_k = sh\theta_{-k} \text{ here})$$

t-dependence of $a_k(t)$ follows from that of $\alpha(t)$ and $\beta(t)$:

$$a_{k'}(t) = ch\theta_{k'}e^{-iE_{k'}t}\alpha_{k'} - sh\theta_{k'}e^{iE_{k'}t}\beta^+_{-k'}$$

$$\Rightarrow [a_{k'}(t), a_k] = 0$$

$$[a_{k'}(t), a^+_{-k}] = ch\theta_{k'}e^{-iE_{k'}t}ch\theta_k\overbrace{[\alpha_{k'}, \alpha^+_{-k}]}^{\delta_{k',-k}}$$

$$+ sh\theta_{k'}e^{iE_{k'}t}sh\theta_k\underbrace{[\beta^+_{-k'}, \beta_k]}_{-\delta_{-k',k}} =$$

$$= ch^2\theta_k \, e^{-iE_k t} - sh^2\theta_k \, e^{iE_k t}$$

Thus, collecting everything

$$\chi_{rr'}(\omega) = -i(2S)\int dt\, e^{i\omega t} \frac{2}{N}\sum_k e^{-i\bar{k}(\bar{r}-\bar{r}')}$$

$$\Big(ch^2\theta_k \, e^{-iE_k t} - sh^2\theta_k \, e^{iE_k t} - \big(ch^2\theta_{+k} \, e^{iE_k t} - sh^2\theta_k \, e^{-iE_k t}\big)\Big)$$

from $[a_{k'}(t), a^+_{-k}]$ from $[a^+_{-k'}(t), a_k]$

$$= -i(2S)\int_0^\infty dt\, e^{i(\omega+i\delta)t} \frac{2}{N}\sum_k e^{ik(r-r')}$$

$$\cdot \Big\{ ch^2\theta_k \left(e^{-iE_k t} - e^{iE_k t}\right)$$

$$+ sh^2\theta_k \left(e^{-iE_k t} - e^{iE_k t}\right) \Big\}$$

$$ch^2\theta_k + sh^2\theta_k = ch(2\theta_k).$$

$$\int_0^\infty dt\, e^{i(\omega+i\delta)t \pm iE_k t} = \int_0^\infty dt\, e^{it(\omega \pm E_k + i\delta)}$$

$$= \frac{1}{i(\omega \pm E_k + i\delta)} \, e^{it(\omega \pm E_k + i\delta)}\Big|_0^\infty = \frac{i}{\omega \pm E_k + i\delta}$$

Since $\delta = +0$, $e^{it(\Omega + i\delta)} = e^{it\Omega} e^{-t\delta} \xrightarrow[t\to\infty]{} 0$

$$\langle S^x(r,t)\rangle = -\frac{i}{\hbar}\int_{-\infty}^{t} dt' \int dr' \langle [S^x(r,t), S^x(r',t')]\rangle h(r')$$

let $\vec{r} \in A$ sublattice.

Since $[a_r, b^+_{r'}] = 0$, we need

$$\langle [(a(r,t) + a^+(r,t)), (a(r',t') + a^+(r',t'))]\rangle$$

Now, $a_r(t) = \sqrt{\frac{2}{N}} \sum_k e^{i\vec{k}\vec{r}} a_k(t) = \sqrt{\frac{2}{N}} \sum_k e^{i\vec{k}\vec{r}} (\mathrm{ch}\,\theta_k\, \alpha_k(t) - \mathrm{sh}\,\theta_k\, \beta^+_{-k}(t))$

The t-dependence follows from (obtained earlier)

$$H^{(2)}_{AFM} = JS \sum_q \omega_q (\alpha^+_q \alpha_q + \beta^+_q \beta_q)$$

$$\alpha_k(t) = e^{iHt}\, \alpha_k\, e^{-iHt}$$

$$\frac{d}{dt}\alpha_k(t) = i\,[H, \alpha_k(t)] = i \sum_q JS\omega_q\, [\alpha^+_q \alpha_q, \alpha_k]$$

$$\alpha^+_q \alpha_q \alpha_k - \alpha_k \alpha^+_q \alpha_q = (\alpha^+_q \alpha_k - \alpha_k \alpha^+_q)\alpha_q = -\delta_{kq}\alpha_q$$

$$= -i \sum_q JS\omega_q\, \delta_{k,q}\, \alpha_q(t) = -i\, JS\omega_k\, \alpha_k(t)$$

$$\Rightarrow \alpha_k(t) = e^{-iE_k t}\, \alpha_k \qquad E_k = E_{-k} = JS\omega_k.$$

Similarly, $\beta^+_{-k}(t) = e^{iE_k t}\, \beta^+_{-k}$

$$\chi_{rr'}(\omega) = -i\,(2S)\,\frac{2}{N}\sum_k e^{i\bar{k}(\bar{r}-\bar{r}')}\,ch(2\theta_k)$$

$$\cdot\left(\frac{i}{\omega - E_k + i\delta} - \frac{i}{\omega + E_k + i\delta}\right) = \frac{2}{N}\sum_k e^{i\bar{k}(\bar{r}-\bar{r}')}\,ch(2\theta_k)\cdot$$

$$\left(\frac{1}{\omega - E_k + i\delta} - \frac{1}{\omega + E_k + i\delta}\right)$$

$$ch\,2\theta_k = \frac{A_k}{\omega_k} = \frac{JSA_k}{E_k} = \frac{JS(2d)}{E_k}$$

Observe ~~too~~ that $\chi_{rr'}(\omega)$ depends on $(\vec{r}-\vec{r}')$ only

$$\Rightarrow \chi_{rr'}(\omega) = \chi(\bar{r}-\bar{r}', \omega)$$

$$\chi(\bar{q},\omega) = \int d\vec{r}\, e^{-i\bar{q}\bar{r}}\,\chi(\vec{r},\omega) =$$

$$= \int d\vec{r}\, e^{-i\bar{q}\bar{r}}\,\frac{2}{N}\sum_k e^{i\bar{k}\bar{r}}\,(\ldots)_{\vec{k}}$$

$$= \frac{2}{N}\sum_k \int d\vec{r}\, e^{i\vec{r}(\vec{k}-\vec{q})}\,(\ldots)_{\vec{k}}$$

$$= (\ldots)_{\vec{q}}$$

$$\left[\text{because } \frac{2}{N}\sum_k \phi_k = \int \frac{d\vec{k}}{(2\pi)^3}\,\phi_k\right]$$

So that finally

$$\chi(\vec{q},\omega) = \frac{6JS}{E_q}\left(\frac{1}{\omega - E_q + i\delta} - \frac{1}{\omega + E_q + i\delta}\right)$$

$$E_q = JS\,\omega_q = JS\sqrt{A_q^2 - B_q^2} \approx c\,|\vec{q}\,|$$

Since $\dfrac{1}{x \pm i0} = P\dfrac{1}{x} \mp i\pi\,\delta(x)$

$$\Rightarrow \operatorname{Im}\chi(\vec{q},\omega) = \frac{6JS}{E_q}(-\pi)\left(\delta(\omega - E_q) - \delta(\omega + E_q)\right)$$

has singularity at $\omega = E_q$ (for $\omega > 0$)
which are excitation energies of spin waves

Rate of energy absorption in AC field

$$H = H(t) = H_0 + H'(t) \Rightarrow E(t) = \mathrm{Tr}\big(\rho(t)\, H(t)\big) \text{ is not conserved!}$$

$$\frac{dE}{dt} = \mathrm{Tr}\left[\rho(t)\frac{dH}{dt}\right] + \mathrm{Tr}\left[\frac{d\rho}{dt} H\right]$$

But $\frac{d\rho}{dt} = -i\,[H(t), \rho(t)] \Rightarrow \mathrm{Tr}\big\{[H,\rho]H\big\} =$

$$= \mathrm{Tr}\{H\rho H - \rho H H\} = \mathrm{Tr}\{H H \rho\} - \mathrm{Tr}\{H H \rho\} = 0$$

$$\Rightarrow \frac{dE}{dt} = \mathrm{Tr}\left\{\rho(t)\frac{dH}{dt}\right\}$$

Consider monochromatic perturbation

$$H'(t) = \hat{O}\, h e^{i\omega t} + \hat{O}^{+} h^{*} e^{-i\omega t}$$

where h, h^{*} are complex amplitudes. Note $H'(t) = (H'(t))^{+}$

Then $\frac{dH}{dt} = i\omega\left(\hat{O} h e^{i\omega t} - \hat{O}^{+} h^{*} e^{-i\omega t}\right)$

$$\Rightarrow \frac{dE}{dt} = i\omega\Big(\underbrace{\mathrm{Tr}\{\rho(t)\hat{O}\}}_{\langle \hat{O}(t)\rangle} h e^{i\omega t} - \underbrace{\mathrm{Tr}\{\rho(t)\hat{O}^{+}\}}_{\langle \hat{O}^{+}(t)\rangle} h^{*} e^{-i\omega t}\Big).$$

Linear response tells us

$$\langle \hat{O}(t)\rangle = \int_{-\infty}^{t} dt' \left(-i\langle[\hat{O}(t), \hat{O}(t')\, h e^{i\omega t'} + \hat{O}^{+}(t')\, h^{*} e^{-i\omega t'}]\rangle\right)$$

$$= \int_{-\infty}^{t} dt' \left(\chi_{OO}(t-t')\, h e^{i\omega t'} + \chi_{OO^{+}}(t-t')\, h^{*} e^{-i\omega t'}\right)$$

It is customary to calculate energy absorption over the period $T = \frac{2\pi}{\omega}$

$$\overline{\frac{dE}{dt}} = \frac{1}{T}\int_0^T dt\, \frac{dE}{dt} = \frac{1}{T}\int_0^{2\pi/\omega} dt\, (i\omega)\Big\{ \langle \hat{O}(t)\rangle h e^{i\omega t}$$

$$- \langle \hat{O}^+(t)\rangle h^* e^{-i\omega t} \Big\} =$$

$$= \frac{i\omega}{T}\int_0^{2\pi/\omega} dt \Big\{ \int_{-\infty}^{t} dt' \left(\chi_{OO}(t-t') h e^{i\omega t'} + \chi_{OO^+}(t-t') h^* e^{-i\omega t'} \right) \cdot$$

$$\cdot\, h e^{i\omega t}$$

$$- \int_{-\infty}^{t} dt' \left(\chi_{O^+O}(t-t') h e^{i\omega t'} + \chi_{O^+O^+}(t-t') h^* e^{-i\omega t'} \right) \cdot$$

$$\cdot\, h^* e^{-i\omega t} \Big\}$$

Change $t' = t - u \Rightarrow \int_{-\infty}^{t} dt'\, \chi(t-t')\, e^{\pm i\omega t'} =$

$$= \int_0^{\infty} du\, \chi(u)\, e^{\pm i\omega(t-u)}$$

$$\overline{\frac{dE}{dt}} = \frac{i\omega}{T}\int_0^{2\pi/\omega} dt \Big\{ \int_0^{\infty} du \left(\chi_{OO}(u)\, h e^{i\omega(t-u)} + h^* e^{-i\omega(t-u)} \chi_{OO^+}(u) \right) \cdot$$

$$\cdot\, h e^{i\omega t}$$

$$- \int du \left(\chi_{O^+O}(u)\, h e^{i\omega(t-u)} + \chi_{O^+O^+}(u)\, e^{-i\omega(t-u)} \right) h^* e^{-i\omega t} \Big\}$$

But $\frac{1}{T}\int_0^{2\pi/\omega} dt\, e^{i2\omega t} = \frac{1}{T \cdot i2\omega}\left(e^{i2\omega \frac{2\pi}{\omega}} - 1 \right) = 0$; $\frac{1}{T}\int_0^{2\pi/\omega} dt = 1.$

So that $\overline{\frac{dE}{dt}} = i\omega \int_0^\infty du \left(\chi_{00+}(u) e^{i\omega u} h^* h \right.$

$$\left. - \chi_{0+0}(u) e^{-i\omega u} h h^* \right) = |h|^2 i\omega \left(\chi_{00+}(\omega) - \chi_{0+0}(-\omega) \right)$$

Here $\chi_{00+}(\omega) = \int_0^\infty du\, e^{i\omega u} \chi_{00+}(u) =$

$$= -i \int_0^\infty du\, e^{i\omega u} \langle [\hat{O}(u), \hat{O}^+] \rangle$$

Then $\chi_{0+0}(-\omega) = -i \int_0^\infty du\, e^{-i\omega u} \langle [\hat{O}^+(u), \hat{O}] \rangle$

Hence $\cancel{\left(\chi_{0+0}(-\omega)\right)^* = i \int_0^\infty du\, e^{i\omega u} \langle [\hat{O}^+(u), \hat{O}] \rangle^*}$

$\cancel{= i \int du\, e^{i\omega u} \langle \rho_0 [e^{iHu} \hat{O}^+ e^{-iHu}, \hat{O}] \rangle^*}$

$$\left(\chi_{00+}(\omega)\right)^* = i \int_0^\infty du\, e^{-i\omega u} \langle \rho_0 [e^{iHu} \hat{O} e^{-iHu}, \hat{O}^+] \rangle^*$$

$$= i \int_0^\infty du\, e^{-i\omega u} \langle \rho_0 [\hat{O}(u), \hat{O}^+]^+ \rangle =$$

$$= i \int_0^\infty du\, e^{-i\omega u} \langle \rho_0 \left(\hat{O}(u) \hat{O}^+ - \hat{O}^+ \hat{O}(u) \right)^+ \rangle =$$

$$= i \int_0^\infty du\, e^{-i\omega u} \langle \rho_0 \left(\hat{O} \hat{O}^+(u) - \hat{O}^+(u) \hat{O} \right) \rangle =$$

$$= -i \int_0^\infty du\, e^{-i\omega u} \langle \rho_0 [\hat{O}^+(u), \hat{O}] \rangle$$

$$= \chi_{0+0}(-\omega)$$

Thus $\overline{\frac{dE}{dt}} = |h|^2(i\omega)\ 2i\ \mathrm{Im}(\chi_{00+}(\omega))$

$$= -2\omega\ \mathrm{Im}(\chi_{00+}(\omega)) \cdot |h|^2.$$

Rate of energy absorption is determined by Im part of the susceptibility!

This is manifestation of FDT (fluctuation-dissipation theorem).

And so, if field $\vec{h}$ couples to excitations with momentum $\vec{q}$

$$\frac{dE}{dt} = -2\omega |h_q|^2 \, \mathrm{Im}\left(\chi_{00+}(\vec{q},\omega)\right) =$$

$$= 2\omega |h_q|^2 \, \frac{6JS\pi}{E_q} \left(\delta(\omega - E_q) - \delta(\omega + E_q)\right)$$

$$= \left(\text{for } \omega > 0 \text{ of course}\right)$$

$$= 12\pi JS \, |h_q|^2$$

* This is the basic idea of NMR, ESR and many other <u>resonanse</u> techniques. Read Ch. 3.2 of Blundell's book.